

Bank Direct Marketing Dataset — Metadata Description

Dataset Business Context

This dataset represents **bank direct marketing campaigns**, where a bank contacts customers (mainly by phone) to convince them to subscribe to a **term deposit**.

The goal of analysis is usually to understand:

- Customer behavior
- Campaign effectiveness
- Factors affecting subscription decisions

Target Variable: y → whether the client subscribed or not.

Bank Direct Marketing Dataset — Detailed Column Metadata (Narrative)

age

This column contains **numeric values (integers)** representing the client's age in years. It describes the customer's demographic group and is used to analyze how different age segments behave and how age influences the probability of subscribing to a term deposit.

job

This column contains **categorical values (text labels)** representing the client's profession, such as *admin*, *technician*, *services*, *management*, *blue-collar*, *student*, etc. It describes the customer's occupation and helps the business identify which job categories are more likely to respond to marketing campaigns.

marital

This column contains **categorical values (text labels)** representing the client's marital status, such as *married*, *single*, or *divorced*. It describes the client's social status and helps understand how personal life situation affects financial decisions.

education

This column contains **categorical values (text labels)** representing the client's level of education, such as *basic*

education, high school, university degree, or unknown. It reflects the customer's knowledge level and may indicate financial awareness or decision-making ability.

default

This column contains **binary categorical values (yes/no)** indicating whether the client has credit in default. It represents the customer's credit risk level and financial reliability.

housing

This column contains **binary categorical values (yes/no)** indicating whether the client has a housing loan. It describes the customer's financial commitments and long-term debt obligations.

loan

This column contains **binary categorical values (yes/no)** indicating whether the client has a personal loan. It reflects the client's current debt situation and financial pressure.

contact

This column contains **categorical values (text labels)** representing the communication method used to contact the client, such as *telephone* or *cellular*. It helps evaluate which communication channel is more effective for marketing.

month

This column contains **categorical values (text labels)** representing the month of the last contact (jan, feb, mar, etc.). It describes when the campaign interaction occurred and helps analyze seasonal effects.

day_of_week

This column contains **categorical values (text labels)** representing the day of the week of the last contact (mon, tue, wed, etc.). It helps determine which days produce better customer responses.

campaign

This column contains **numeric values (integers)** representing the number of times the client was contacted during the current marketing campaign. It measures the level of marketing effort applied to each customer.

pdays

This column contains **numeric values (integers)** representing the number of days since the client was last contacted in a previous campaign. Special values like *999* or *-1* indicate that the client was never contacted before. It measures customer recency and engagement history.

previous

This column contains **numeric values (integers)** representing the number of contacts made with the client before the current campaign. It reflects the historical relationship between the bank and the customer.

poutcome

This column contains **categorical values (text labels)** representing the outcome of the previous marketing campaign, such as *success*, *failure*, or *unknown*. It helps evaluate past campaign performance and customer responsiveness.

emp.var.rate

This column contains **numeric values (decimal numbers)** representing the employment variation rate (economic indicator). It describes changes in employment levels in the economy and reflects the overall economic environment.

cons.price.idx

This column contains **numeric values (decimal numbers)** representing the consumer price index. It measures inflation and cost of living changes affecting customer financial behavior.

cons.conf.idx

This column contains **numeric values (decimal numbers)** representing the consumer confidence index. It describes how confident customers feel about the economy and their financial future.

euribor3m

This column contains **numeric values (decimal numbers)** representing the 3-month Euribor interest rate. It reflects market interest rates and directly affects deposit attractiveness.

nr.employed

This column contains **numeric values (decimal numbers)** representing the number of employees in the

economy. It is an economic indicator that reflects overall economic stability.

y (Target Variable)

This column contains **binary categorical values (yes/no)** indicating whether the client subscribed to a term deposit. This is the main outcome variable used for prediction and business decision-making.